《Ai Agent》第3-21节：在云服务器部署上线

来自： 码农会锁

小傅哥 2025年10月07日 08:15

本章重点 ：★★☆☆☆
课程视频 ：https://t.zsxq.com/wfgDS
代码分支 ：
https://gitcode.net/KnowledgePlanet/ai-agent-station-study/-/tree/tag-v1.0
https://gitcode.net/KnowledgePlanet/ai-agent-station-front/-/tree/tag-v1.0
工程代码 ：
https://gitcode.net/KnowledgePlanet/ai-agent-station-study
https://gitcode.net/KnowledgePlanet/ai-agent-station-front

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> 沉淀、分享、成长，让自己和他人都能有所收获！😄

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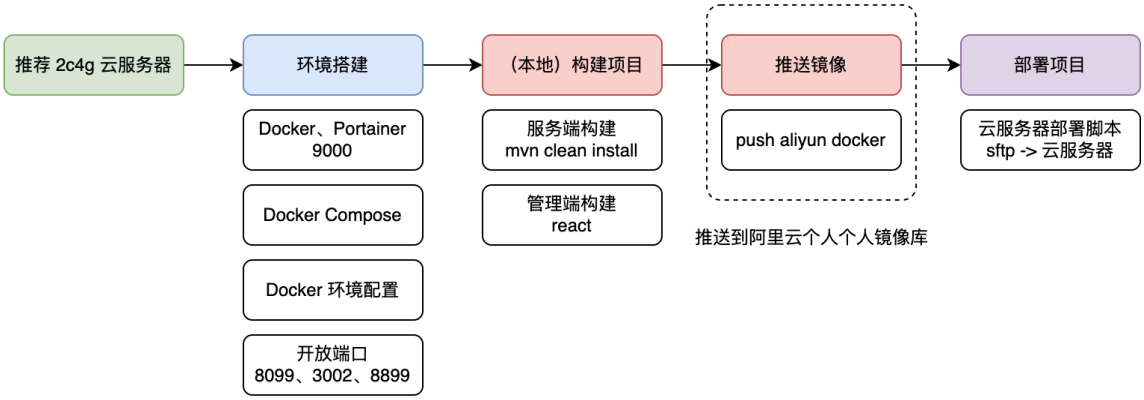
一、本章诉求

在云服务器（2c4g）部署 Ai Agent 前端页面（html）、管理页面（react）、后端服务（java）、基础环境（MySql、PG向量库）。

二、部署过程

如图，为本次的部署过程



推荐2c4g云服务器购买，http://618.gaga.plus/ 选个便宜的购买就可以。

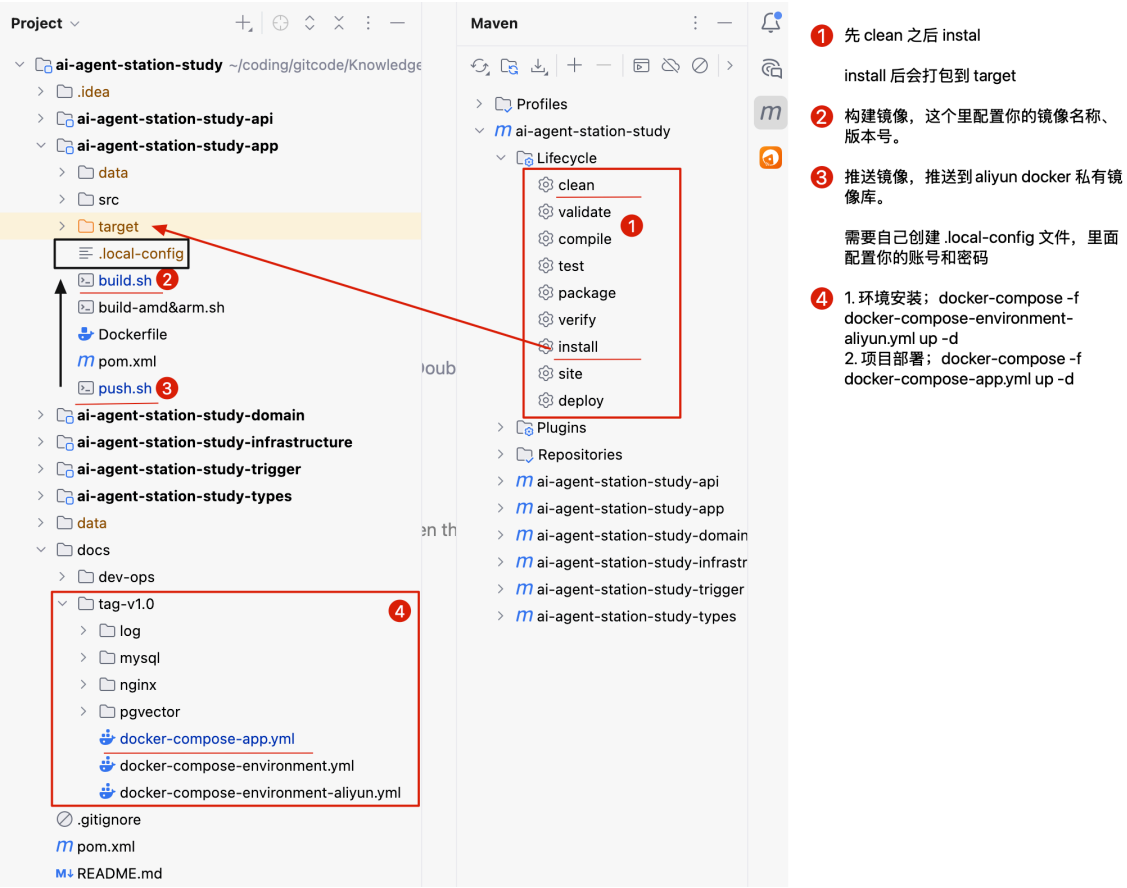
本次会进行本地构建镜像，之后 push 到阿里云私有个人镜像库（免费的可以申请）。这个可以代替 Docker Hub 作为代理使用。教程；https://t.zsxq.com/XdoWr

云服务器部署教程：https://t.zsxq.com/19osWS4qj – 需要云服务器安装 Docker、Portainer，以及配置镜像地址 https://t.zsxq.com/2DGGY

记得在云服务器安全组开放端口 9000（Portainer）、8099（服务端）、3002（管理端）、8899（mysql管理端） 等你需要对外的端口。

三、工程结构

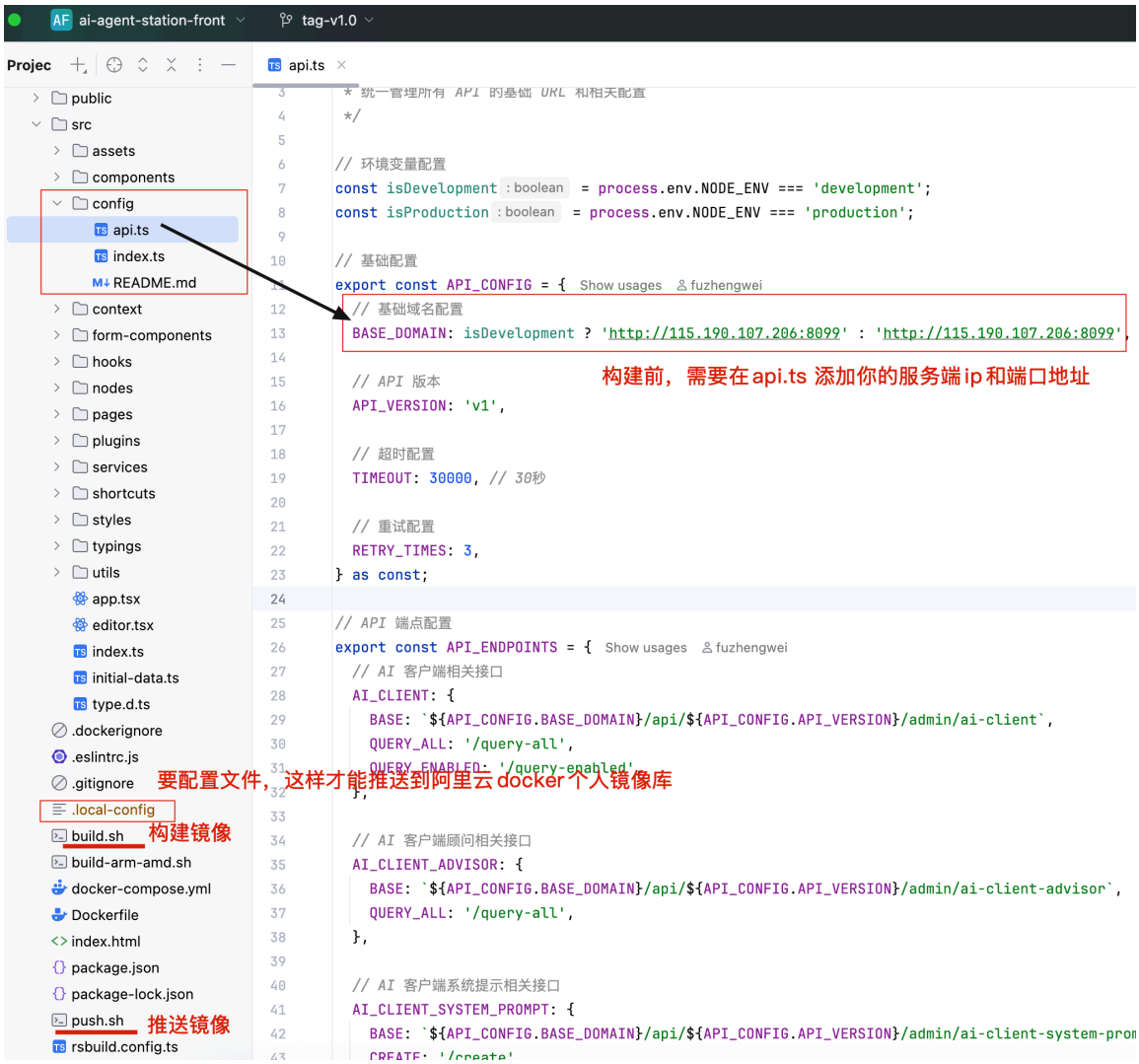
1. 服务端（含用户端）



注意，先参考教程 https://t.zsxq.com/XdoWr 在阿里云创建 docker 私有镜像库，获得账户和密码，配置到(文件也需要自己创建) .local-config

构建整个工程，分为1步 ~ 4步，前面3步是构建和上传镜像，第4步，是部署项目。课程里的第4步会在云服务器操作。

2. 管理端



管理端，主要需要修改下 api.ts 的 IP 地址。修改后再构建。在公司里，一般服务端都是有一个固定的 API 网关的域名地址，那么就不需要修改了，统一走一套网关的域名地址，转换到对应要调用的服务即可。

`.local-config` 文件还是需要自己创建，里面需要填写的信息 push.sh 里有说明。如果未创建，在执行 `./push.sh` 也会提示。

以上配置好后，依次执行 `./build.sh` 和 `./push.sh` 进行镜像的构建和推送。

四、上云部署

1. 镜像构建

1.1 服务端

```
fuzhengwei@192 ai-agent-station-study-app % /bin/zsh /Users/fuzhengwei/coding/gitcode/KnowledgePla
[+] Building 3.8s (11/11) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 1.41kB
 => [internal] load metadata for registry.cn-hangzhou.aliyuncs.com/xfg-studio/openjdk:17-ea-17-jdk
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM registry.cn-hangzhou.aliyuncs.com/xfg-studio/openjdk:17-ea-17-jdk-slim-buster@sha25
 => => resolve registry.cn-hangzhou.aliyuncs.com/xfg-studio/openjdk:17-ea-17-jdk-slim-buster@sha25
 => [internal] load build context
 => => transferring context: 161B
 => CACHED [2/5] RUN ln -snf /usr/share/zoneinfo/PRC /etc/localtime && echo PRC > /etc/timezone
 => CACHED [3/5] RUN echo "deb http://archive.debian.org/debian buster main" > /etc/apt/sources.li
 => CACHED [4/5] RUN node -v && npm -v
 => CACHED [5/5] ADD target/ai-agent-station-study-app.jar /ai-agent-station-study-app.jar
 => exporting to oci image format
 => => exporting layers
 => => exporting manifest sha256:563485714efec04c423b4edf7d4daa2b0b37df564094330b4d730bd97a2a8663
```

```
 => => exporting config sha256:455e938e118b7df6776b39884ebf9beb598a312d58fcea0729cf0934eabb0b98
 => => sending tarball
 => importing to docker

View build details: docker-desktop://dashboard/build/moyu-builder/moyu-builder0/wlftpmrbqkn9xxak87

 1 warning found (use docker --debug to expand):
 - InvalidBaseImagePlatform: Base image registry.cn-hangzhou.aliyuncs.com/xfg-studio/openjdk:17-ea
fuzhengwei@192 ai-agent-station-study-app % /bin/bash /Users/fuzhengwei/coding/gitcode/KnowledgePl
Logging into Aliyun Docker Registry...
WARNING! Using --password via the CLI is insecure. Use --password-stdin.
Login Succeeded
Tagging the Docker image...
Pushing the Docker image to Aliyun...
The push refers to repository [registry.cn-hangzhou.aliyuncs.com/fuzhengwei/ai-agent-station-study
4f4fb700ef54: Layer already exists
faf4c47c8c61: Layer already exists
b747e468795b: Layer already exists
acd6b8894973: Pushed
f7ec5a41d630: Layer already exists
86151f1b2bee: Layer already exists
a0fba1dab33c: Layer already exists
1.1: digest: sha256:563485714efec04c423b4edf7d4daa2b0b37df564094330b4d730bd97a2a8663 size: 1630
Docker image pushed successfully!
检出地址：docker pull registry.cn-hangzhou.aliyuncs.com/fuzhengwei/ai-agent-station-study-app:1.1
标签设置：docker tag registry.cn-hangzhou.aliyuncs.com/fuzhengwei/ai-agent-station-study-app:1.1 fuz
Logging out from Aliyun Docker Registry...
Removing login credentials for registry.cn-hangzhou.aliyuncs.com
```

执行了构建脚本，并上传镜像到阿里云 Docker 镜像库。

1.2 管理端

```
fuzhengwei@fuzhengweideMacBook-Pro-2 ai-agent-station-front % /bin/zsh /Users/fuzhengwei/coding/gi
[+] Building 6.3s (20/20) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 2.95kB
 => [internal] load metadata for registry.cn-hangzhou.aliyuncs.com/xfg-studio/nginx:1.25.1
 => [internal] load metadata for registry.cn-hangzhou.aliyuncs.com/xfg-studio/node:20-alpine
 => [internal] load .dockerignore
 => => transferring context: 874B
 => [runner 1/5] FROM registry.cn-hangzhou.aliyuncs.com/xfg-studio/nginx:1.25.1@sha256:73e957703f1
 => => resolve registry.cn-hangzhou.aliyuncs.com/xfg-studio/nginx:1.25.1@sha256:73e957703f1266530d
 => [internal] load build context
 => => transferring context: 45.14kB
 => [base 1/1] FROM registry.cn-hangzhou.aliyuncs.com/xfg-studio/node:20-alpine@sha256:1d0029b1aa9
 => => resolve registry.cn-hangzhou.aliyuncs.com/xfg-studio/node:20-alpine@sha256:1d0029b1aa9dcb9c
 => CACHED [builder 1/4] WORKDIR /app
 => CACHED [deps 1/5] RUN apk add --no-cache libc6-compat
 => CACHED [deps 2/5] WORKDIR /app
 => CACHED [deps 3/5] COPY package.json yarn.lock* package-lock.json* pnpm-lock.yaml* ./
 => CACHED [deps 4/5] RUN yarn config set registry 'https://registry.npmmirror.com/'
 => CACHED [deps 5/5] RUN yarn install
 => CACHED [builder 2/4] COPY --from=deps /app/node_modules ./node_modules
 => [builder 3/4] COPY . .
 => [builder 4/4] RUN npx rsbuild build
 => CACHED [runner 2/5] WORKDIR /usr/share/nginx/html
 => CACHED [runner 3/5] RUN rm -rf ./*
 => [runner 4/5] COPY --from=builder /app/dist .
 => [runner 5/5] RUN echo 'server {' > /etc/nginx/conf.d/default.conf &&     echo '    listen 3002
WARNING: No output specified with docker-container driver. Build result will only remain in the bu
```

```
View build details: docker-desktop://dashboard/build/moyu-builder/moyu-builder0/g23le8qcwwygx9kfti

 2 warnings found (use docker --debug to expand):
 - InvalidBaseImagePlatform: Base image registry.cn-hangzhou.aliyuncs.com/xfg-studio/nginx:1.25.1
 - InvalidBaseImagePlatform: Base image registry.cn-hangzhou.aliyuncs.com/xfg-studio/node:20-alpin
fuzhengwei@fuzhengweideMacBook-Pro-2 ai-agent-station-front % /bin/zsh /Users/fuzhengwei/coding/gi
[+] Building 7.0s (22/22) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 2.95kB
 => [internal] load metadata for registry.cn-hangzhou.aliyuncs.com/xfg-studio/node:20-alpine
 => [internal] load metadata for registry.cn-hangzhou.aliyuncs.com/xfg-studio/nginx:1.25.1
 => [internal] load .dockerignore
 => => transferring context: 874B
 => [internal] load build context
 => => transferring context: 41.13kB
 => [base 1/1] FROM registry.cn-hangzhou.aliyuncs.com/xfg-studio/node:20-alpine@sha256:1d0029b1aa9
 => => resolve registry.cn-hangzhou.aliyuncs.com/xfg-studio/node:20-alpine@sha256:1d0029b1aa9dcb9c
 => [runner 1/5] FROM registry.cn-hangzhou.aliyuncs.com/xfg-studio/nginx:1.25.1@sha256:73e957703f1
 => => resolve registry.cn-hangzhou.aliyuncs.com/xfg-studio/nginx:1.25.1@sha256:73e957703f1266530d
 => CACHED [builder 1/4] WORKDIR /app
 => CACHED [deps 1/5] RUN apk add --no-cache libc6-compat
 => CACHED [deps 2/5] WORKDIR /app
 => CACHED [deps 3/5] COPY package.json yarn.lock* package-lock.json* pnpm-lock.yaml* ./
 => CACHED [deps 4/5] RUN yarn config set registry 'https://registry.npmmirror.com/'
 => CACHED [deps 5/5] RUN yarn install
 => CACHED [builder 2/4] COPY --from=deps /app/node_modules ./node_modules
 => [builder 3/4] COPY . .
 => [builder 4/4] RUN npx rsbuild build
 => CACHED [runner 2/5] WORKDIR /usr/share/nginx/html
 => CACHED [runner 3/5] RUN rm -rf ./*
 => CACHED [runner 4/5] COPY --from=builder /app/dist .
 => CACHED [runner 5/5] RUN echo 'server {' > /etc/nginx/conf.d/default.conf &&     echo '    list
 => exporting to oci image format
 => => exporting layers
 => => exporting manifest sha256:a8ebca96949be2170c7cd4635a0b2781959863aa95920dba8068c70d0d843a2f
 => => exporting config sha256:90af0263611b1488b6cccdf58f66ebac4c3181f2b6d9080da447b894ffbcba08
 => => sending tarball
 => importing to docker

View build details: docker-desktop://dashboard/build/moyu-builder/moyu-builder0/mtv9tdbxovede54pc0

 2 warnings found (use docker --debug to expand):
 - InvalidBaseImagePlatform: Base image registry.cn-hangzhou.aliyuncs.com/xfg-studio/node:20-alpin
 - InvalidBaseImagePlatform: Base image registry.cn-hangzhou.aliyuncs.com/xfg-studio/nginx:1.25.1
fuzhengwei@fuzhengweideMacBook-Pro-2 ai-agent-station-front % /bin/bash /Users/fuzhengwei/coding/g
Logging into Aliyun Docker Registry...
WARNING! Using --password via the CLI is insecure. Use --password-stdin.
2025/10/05 20:55:18 notifying Desktop of credentials store update: Post "http://ipc/registry/creds
Login Succeeded
Tagging the Docker image...
Pushing the Docker image to Aliyun...
The push refers to repository [registry.cn-hangzhou.aliyuncs.com/fuzhengwei/ai-agent-station-admin
4f4fb700ef54: Mounted from fuzhengwei/joyagent-jdgenie
e66d0270d23f: Pushed
cbf42f5a00d2: Pushed
8015f365966b: Pushed
a812a0b054a2: Pushed
648e0aadf75a: Pushed
3459d4f97715: Pushed
55ac49bd649c: Pushed
9f91ccd92daa: Pushed
4cadff8bc2aa: Pushed
262696647b70: Pushed
1.0: digest: sha256:a8ebca96949be2170c7cd4635a0b2781959863aa95920dba8068c70d0d843a2f size: 2372
```

```
Docker image pushed successfully!
检出地址：docker pull registry.cn-hangzhou.aliyuncs.com/fuzhengwei/ai-agent-station-admin-app:1.0
标签设置：docker tag registry.cn-hangzhou.aliyuncs.com/fuzhengwei/ai-agent-station-admin-app:1.0 fuz
Logging out from Aliyun Docker Registry...
Removing login credentials for registry.cn-hangzhou.aliyuncs.com
```

执行了构建脚本，并上传镜像到阿里云 Docker 镜像库。

2. 服务部署

2.1 部署脚本

```
# /usr/local/bin/docker-compose -f /docs/dev-ops/environment/environment-docker-compose-2.4.yml up
version: '3.8'
# docker-compose -f docker-compose-app.yml up -d
services:
  # 部署前端项目
  ai-agent-station-front-app:
    image: registry.cn-hangzhou.aliyuncs.com/xfg-studio/nginx:latest
    container_name: ai-agent-station-front-app
    restart: always
    ports:
      - '80:80'
    volumes:
      - ./nginx/html:/usr/share/nginx/html
    privileged: true
    networks:
      - my-network

  # 管理后台
  ai-agent-station-admin-app:
#    image: fuzhengwei/ai-agent-station-admin-app:1.0
    image: registry.cn-hangzhou.aliyuncs.com/fuzhengwei/ai-agent-station-admin-app:1.0
    container_name: ai-agent-station-admin-app
    ports:
      - "3002:3002"
    restart: unless-stopped
    networks:
      - my-network
    healthcheck:
      test: [ "CMD", "curl", "-f", "http://localhost:3002" ]
      interval: 30s
      timeout: 10s
      retries: 3
      start_period: 40s
    labels:
      - "traefik.enable=true"
      - "traefik.http.routers.ai-agent-front.rule=Host(`localhost`)"
      - "traefik.http.services.ai-agent-front.loadbalancer.server.port=3002"

  # 进入后端命令 docker exec -it ai-agent-station-study-app /bin/bash
  ai-agent-station-study-app:
#    image: fuzhengwei/ai-agent-station-study-app:1.1
    image: registry.cn-hangzhou.aliyuncs.com/fuzhengwei/ai-agent-station-study-app:1.1
    container_name: ai-agent-station-study-app
    restart: on-failure
    ports:
      - "8099:8099"
    environment:
      - TZ=PRC
      - SERVER_PORT=8099
      - SPRING_DATASOURCE_MYSQL_USERNAME=root
      - SPRING_DATASOURCE_MYSQL_PASSWORD=123456
```

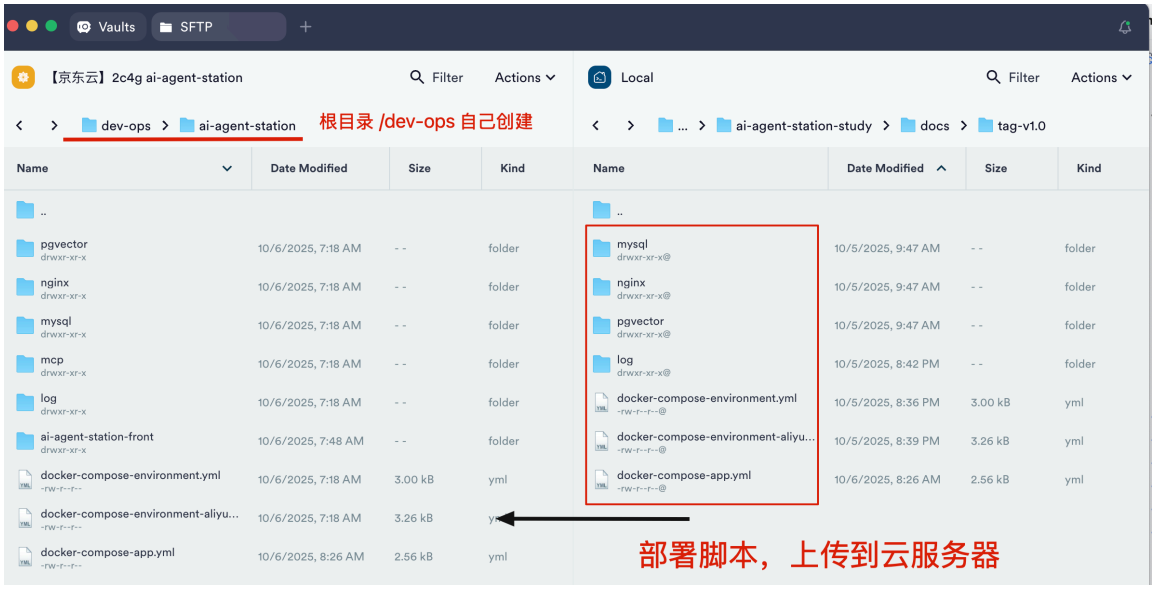
```
      - SPRING_DATASOURCE_MYSQL_URL=jdbc:mysql://mysql:3306/ai-agent-station-study?serverTimezone=
      - SPRING_DATASOURCE_MYSQL_DRIVER_CLASS_NAME=com.mysql.cj.jdbc.Driver
      - SPRING_DATASOURCE_PGVECTOR_DRIVER_CLASS_NAME=org.postgresql.Driver
      - SPRING_DATASOURCE_PGVECTOR_USERNAME=postgres
      - SPRING_DATASOURCE_PGVECTOR_PASSWORD=postgres
      - SPRING_DATASOURCE_PGVECTOR_URL=jdbc:postgresql://vector_db:5432/ai-rag-knowledge?connectTi
      - SPRING_AI_OPENAI_BASE_URL=https://apis.itedus.cn
      - SPRING_AI_OPENAI_API_KEY=sk-Vb7vnKNtGFAsKGhz6765DfE3F1974908Bb66**** 可以联系小傅哥获得申请渠道
      - SPRING_AI_OPENAI_AGENT_AUTO-CONFIG_ENABLED=true
    volumes:
      - ./log:/data/log
    logging:
      driver: "json-file"
      options:
        max-size: "10m"
        max-file: "3"
    networks:
      - my-network

networks:
  my-network:
    driver: bridge
```

ai-agent-station-front-app，是通过 nginx 部署的前端 html 页面。

ai-agent-station-admin-app，是部署的管理后台，注意这个镜像地址要更换为你的 `registry.cn-hangzhou.aliyuncs.com/fuzhengwei/ai-agent-station-admin-app:1.0`

ai-agent-station-study-app，是部署的服务端，这里要注意，SPRING_AI_OPENAI_BASE_URL、SPRING_AI_OPENAI_API_KEY 要更换为你的。

2.2 上传脚本



通过 https://termius.com/ 工具（免费）的 sftp 能力，把脚本上传到云服务器。

2.3 执行脚本

2.3.1 环境安装

```
[root@iv-ydw2iok0lcbw80bxaha0 dev-ops]# cd ai-agent-station/
[root@iv-ydw2iok0lcbw80bxaha0 ai-agent-station]# ls
ai-agent-station-front  docker-compose-environment-aliyun.yml  log  mysql  pgvector
docker-compose-app.yml  docker-compose-environment.yml         mcp  nginx
[root@iv-ydw2iok0lcbw80bxaha0 ai-agent-station]# docker-compose -f docker-compose-environment-aliy
WARN[0000] Found orphan containers ([ai-agent-station-front-app ai-agent-station-study-app ai-agen
[+] Running 4/4
```

```
 ✔ Container mysql            Healthy
 ✔ Container phpmyadmin       Running
 ✔ Container vector_db        Running
 ✔ Container vector_db_admin  Running
[root@iv-ydw2iok0lcbw80bxaha0 ai-agent-station]#
```

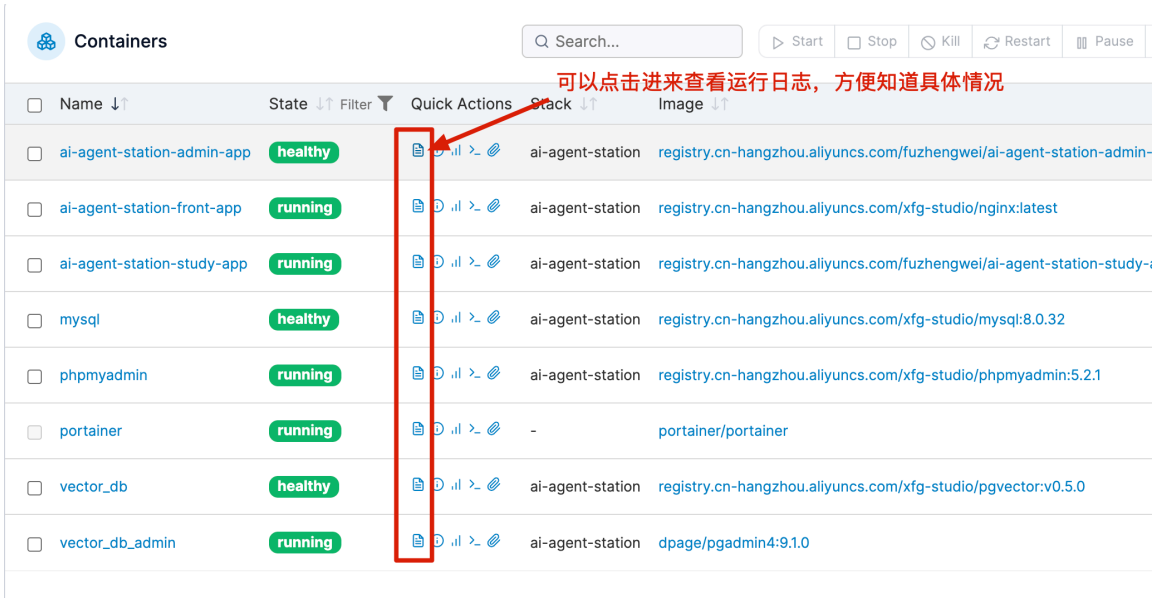
在同一个网络下，安装，系统所需的环境。mysql、vector_db

2.3.2 后端部署 + toc用户端 + 管理端

```
[root@iv-ydw2iok0lcbw80bxaha0 ai-agent-station]# docker-compose -f docker-compose-app.yml up -d
WARN[0000] Found orphan containers ([phpmyadmin vector_db_admin vector_db mysql]) for this project
[+] Running 3/0
 ✔ Container ai-agent-station-admin-app  Running
 ✔ Container ai-agent-station-study-app  Running
 ✔ Container ai-agent-station-front-app  Running
[root@iv-ydw2iok0lcbw80bxaha0 ai-agent-station]#
```

部署项目，包括；服务端（study）、toc用户端（front）、管理端（admin）

2.4 日志查看

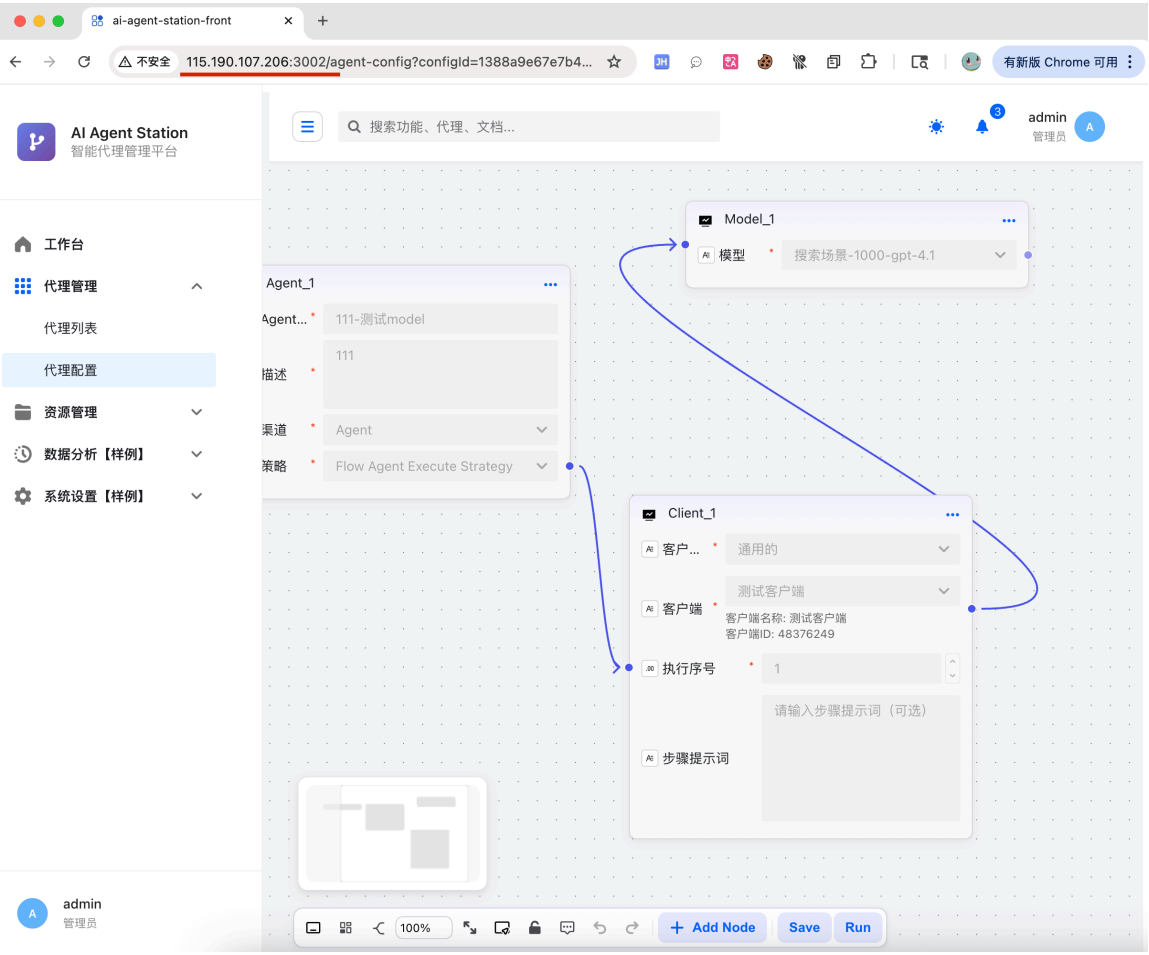


地址：`http://115.190.107.*:9000/#!/2/docker/containers` 进入到你的云服务器 portainer 管理后台地址。

这里可以查看日志，端口，配置。可以多点击下进行了解。

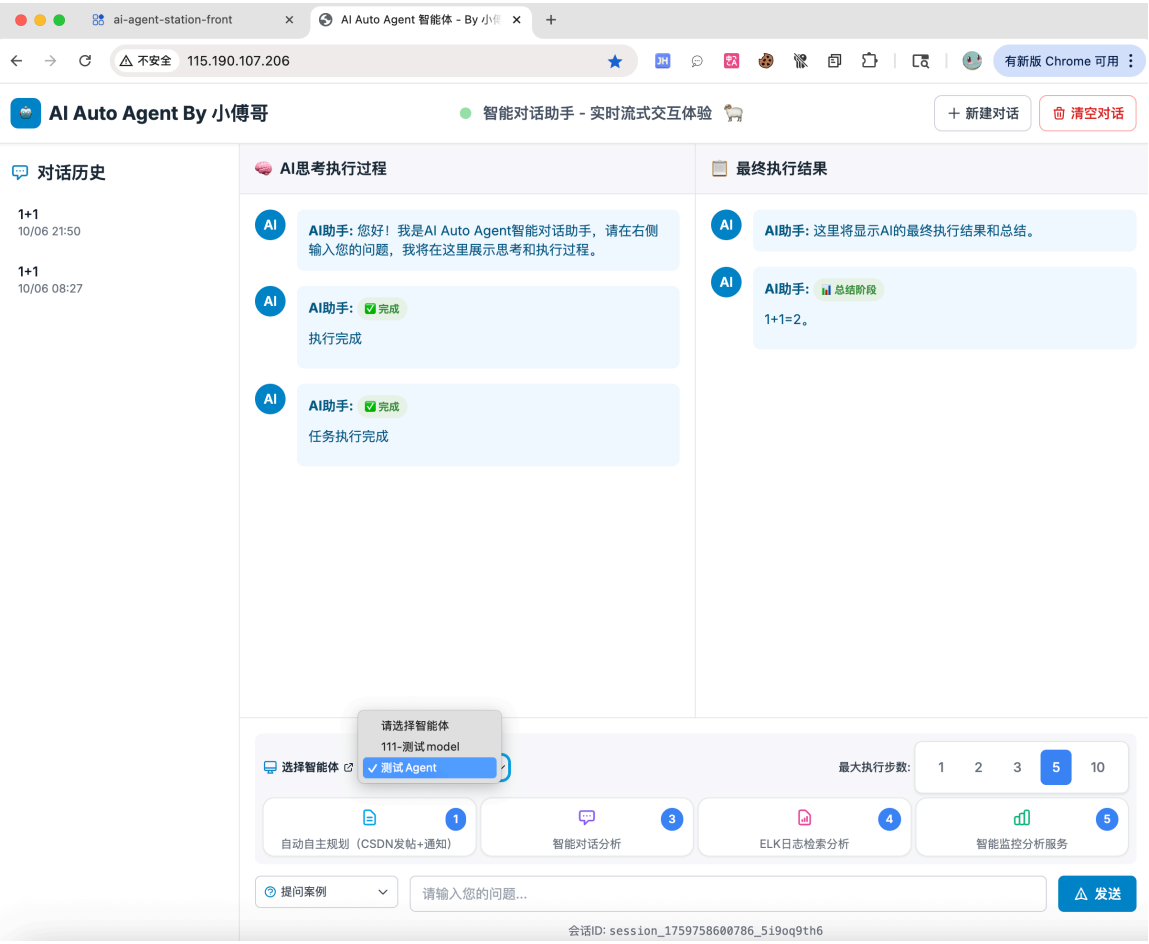
3. 访问服务

3.1 管理端



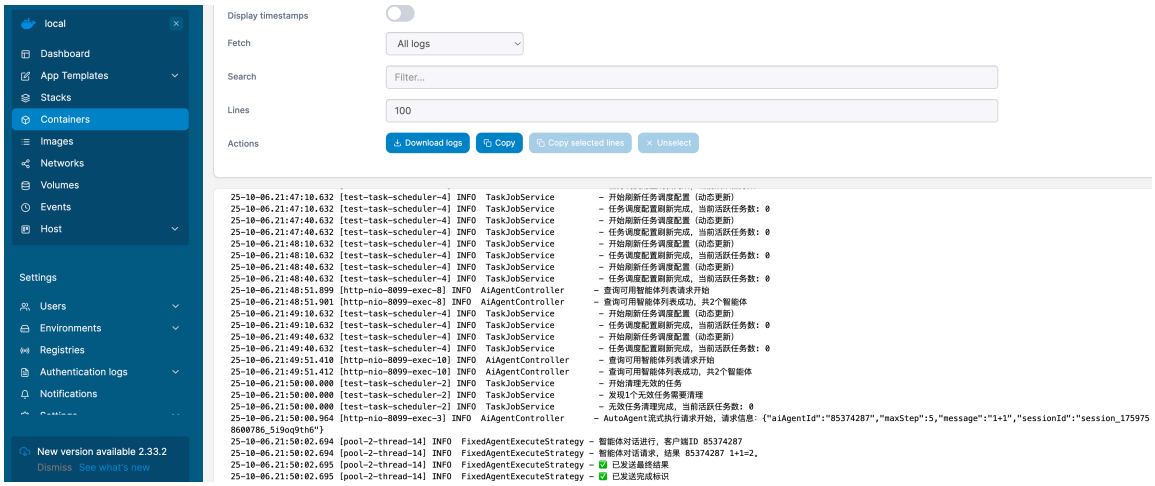
地址：http://115.190.107.*:3002/ 你需要更换为你的服务器ip地址，记得开放 3002 端口。

3.2 用户端



前端对用户的，走的是 80 端口。直接访问ip地址即可。`http://115.190.107.*/`

4. 运行日志



对于运行情况。你可以进入服务端，检查运行日志。

此外，https://bugstack.cn/md/road-map/road-map.html 编程路书，还有很多运营技巧的服务配置，可以参考以及配置使用。

知识星球